HCD Artifacts

River Hawk Consulting, LLC · IMAX HCD Prototype OT · Deliverable ①

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CONTENTS

UNCLASSIFIED. Fabricated composite personas and demonstration content for the prototype's human-centered design baseline, derived from the Phase 2 solution demonstration and its evaluated workflows. No real persons or positions are described.

Built-state evidence for every annotated screen — live screenshots, test results, and the parity baseline against the evaluated demo — accompanies this delivery under `project/evidence/`.

The Artifacts in This Set

Rendered as a single document for delivery: `../project/delivery/hcd_artifacts.pdf`, regenerated by `./scripts/make-hcd-pdf.sh` after any edit here.

Against TDD Task 2: **2.1** personas and empathy maps · **2.2** task flows and journey map · **2.3** wireframes · **2.4** validation and traceability.

- `personas.md` — four composite personas from the evaluated Phase 2 workflows (2.1)
- `empathy_maps.md` — say / think / do / feel per persona, pains removed, gains delivered (2.1)
- `journey_map.md` — the end-to-end triage journey, the four task flows inside it, and the twenty per-persona success criteria QA replays (2.2)
- `traceability_matrix.md` — every persona pain joined to the design response, the screen, the specification that validates it, and the built-state screenshot (2.4)
- `assumption_log.md` — what we assumed and why, what it costs if wrong, and how to close it; sixteen assumptions, six closed (2.4)
- `linguist_workflow_model.md` — HCD finding (31 Jul): the linguist's unit of gold is the whole thread, not a snippet; design response is the additive thread-level gold flow, to be validated at touchpoint #1
- `bilingual_display_model.md` — HCD finding (31 Jul): no verdict without the source — a native speaker always sees original + translation together; the source line stays on screen in translated view and during correction
- `one_output_model.md` — HCD finding (1 Aug): **this application is a linguist's bench** — one output (promoted thread gold), thread-level enrichment, evidence clips removed; other roles consume the linguist's outputs downstream
- `wireframes.md` — annotated three-panel wireframes, persona-indexed, including the 31 Jul thread-gold additions; offered to the Sponsor at touchpoint #1 (Mon 3 Aug)

Built-state evidence for every annotated screen (live screenshots, test results, parity baseline vs the evaluated demo): `project/evidence/` (accompanying this delivery).

Personas

UNCLASSIFIED — Fabricated composite personas for the prototype's human-centered design baseline. Derived from the Phase 2 solution demonstration and its evaluated workflows; no real persons or positions are described.

P1 — Collection Management Officer ("Dana")

Role. Prioritizes what the exploitation team works next; answers "what did we get and is it worth anyone's time?"

Goals. See at a glance how much of the take is new versus worked; steer linguist hours toward threads with triage signal (watchlist hits, geo-fence activity, attached documents).

Frustrations today. Volume statistics live in one tool, content in another; no cheap way to tell an empty channel from a hot one without opening it.

Prototype touchpoints. Queue/all toggle and new/worked counts; content-type lanes (message / transcript / document); facet counts as corpus-level triage signal; disposition markers (reviewed / flagged / discarded) as the team's shared state.

Success looks like. The morning stand-up runs off the queue counts, not a spreadsheet.

P2 — Language Officer ("Marisol")

Role. Works the queue in her scoped languages; renders judgment on machine translation.

Goals. Move through unworked threads fast; read mixed RTL/LTR without fighting the layout; confirm or correct machine translation and have that verdict travel with the text.

Frustrations today. MT output pasted into documents loses its provenance and her corrections; RTL text breaks in generic viewers; retyping context for every excerpt.

Prototype touchpoints. My-languages scope chips; translate-in-place with linguist-confirmed / linguist-edited verdict badges; RTL/mixed-script rendering; analyst notes attached to messages; clip-to-gold-copy carrying her verdict and provenance.

Success looks like. Her correction is made once, badges the message, and lands in the export exactly as she wrote it.

P3 — Subject-Matter Exploiter (“Ken”)

Role. Deep-reads the substance across languages once triage surfaces a thread; builds the analytic picture from content, documents, and summaries.

Goals. Search content and translations in one query; jump from a hit straight to the evidence in context; pull document text out of images; assemble an evidence trail without leaving the viewer.

Frustrations today. Search that stops at metadata; attachments that require a second tool; losing the thread context when quoting a single message.

Prototype touchpoints. Content-mode search that also matches English translations (“matched in translation” badge); scroll-to-hit with flash; on-demand thread summaries; OCR viewer with block-level extraction and English gloss; officer tags feeding back into triage facets.

Success looks like. Query → hit → thread → clipped evidence in under a minute, provenance intact.

P4 — Targeting Officer (“Priya”)

Role. Follows selectors and entities — people, numbers, places — across the take; maintains watchlists and geographic areas of interest.

Goals. Ask “who talked about this place” and “did any flagged number appear this week” directly; pivot from an entity mention to every other appearance.

Frustrations today. Selector lists checked by hand against exports; geographic questions answered by keyword guesswork.

Prototype touchpoints. Entity-mode search with type restriction (person / geo / selector); geo-fence and watchlist groups as one-click scopes; entity chips with confidence; date-range narrowing; export with the entity provenance line.

Success looks like. A watchlist hit is a search result, not a discovery made three weeks later.

Empathy Maps

UNCLASSIFIED — Companion to `personas.md`. One map per persona: what they say, think, do, and feel while working the take, plus the pains the prototype removes and the gains it delivers. Fabricated composites.

Dana — Collection Management Officer

Says	”How much of yesterday’s take is still untouched?” · “Which channels justify a linguist this week?”
Thinks	Volume without signal is noise; my credibility rides on pointing people at the right threads.
Does	Scans queue counts each morning; reassigns scope chips; checks facet counts before tasking.
Feels	Pressure to justify prioritization; relief when the numbers are on the same screen as the content.

Pains removed: cross-tool reconciliation of stats vs content; opening threads just to see if they’re worked. **Gains:** queue/all counts, disposition markers, and content-type lanes make tasking defensible at a glance.

Marisol — Language Officer

Says	”The MT is close but the second clause is wrong.” · “Which of these are actually mine?”
Thinks	If my correction doesn’t travel with the text, someone will quote the bad MT.
Does	Filters to her languages; toggles translate-in-place; edits and confirms; drops notes for the exploiter.
Feels	Ownership of linguistic judgment; annoyance at layout that mangles RTL; trust when her badge is visible downstream.

Pains removed: provenance-free MT pastes; RTL breakage; re-translating already-worked messages. **Gains:** verdict badges (linguist-confirmed / linguist-edited) that persist into clips and exports.

Ken — Subject-Matter Exploiter

Says	”Where else does this warehouse come up?” · “What’s in the attachment?”
Thinks	Context is the analysis — a quote without its thread is a liability.
Does	Runs content searches across originals and translations; jumps to hits; summarizes long threads; OCRs documents; clips evidence.
Feels	Flow when hit → thread → clip chains without tool switches; suspicion of any excerpt he can’t trace.

Pains removed: metadata-only search; second tool for attachments; hand-built evidence trails. **Gains:** matched-in-translation hits, scroll-to-hit flash, OCR gloss, one-click clips with full provenance.

Priya — Targeting Officer

Says	”Any flagged numbers in the last 48 hours?” · “Scope that to the Khorasan corridor.”
Thinks	Selectors age fast; a list checked weekly is a list checked too late.
Does	Runs entity-mode sweeps; applies watchlist/geo-fence groups; narrows by date; pivots on entity chips.
Feels	Urgency; confidence when a group query returns zero and she can trust the zero.

Pains removed: manual selector checks against exports; keyword guesswork for geography. **Gains:** groups as first-class one-click scopes; typed entity search; provenance-carrying entity clips.

Task Flows & Journey Map

UNCLASSIFIED — Companion to `personas.md` and `empathy_maps.md`. The end-to-end triage journey the prototype supports, the task flow each persona runs inside it, and the success criteria QA replays. Fabricated composites; no real persons, positions or operations.

TDD subtask 2.2. Success criteria here are the ones re-run as executable specifications in `tests/ng/` — this document and the test suite are two views of the same claim, which is what stops it becoming aspirational.

The journey — one thread, four hands

The take arrives; a thread leaves as reviewed gold. Four roles touch it, in this order, and the prototype's job is to make each handoff carry its own evidence rather than requiring the next person to reconstruct it.

```
    TAKE ARRIVES
         │
         ▼
   ┌────────────────┐    what is new, what is worth a linguist's hour?
   │ 1. PRIORITIZE  │    P1 Dana — Collection Management Officer
   └────────────────┘    queue/all counts · content-type lanes · facet counts
           │             —— hands over: a scoped, defensible work queue
           ▼
   ┌────────────────┐    is the machine translation right, and whose judgment says
so?
   │ 2. RENDER      │    P2 Marisol — Language Officer
   │    JUDGMENT    │    my-languages scope · bilingual view · confirm / correct
   └────────────────┘    —— hands over: a thread whose text carries a linguist's
verdict
           │
           ▼
   ┌────────────────┐    what does it mean, and can I prove it?
   │ 3. EXPLOIT     │    P3 Ken — Subject-Matter Exploiter
   └────────────────┘    content search · scroll-to-hit · summary · OCR · notes
           │             —— hands over: substance tied to its source
           ▼
   ┌────────────────┐    who and what, and where else does it appear?
   │ 4. TARGET      │    P4 Priya — Targeting Officer
   └────────────────┘    entity search · watchlist / geo-fence groups · date
narrowing
           │             —— hands over: a selector hit that is a search result, not
a surprise
           ▼
    PROMOTED THREAD GOLD — one output, carrying verdicts, notes and provenance
```

The loop is not strictly linear. Ken's officer tags feed back into Dana's triage facets, and Priya's entity sweeps re-enter the queue at step 1. The prototype treats that back-pressure as normal: disposition markers and tags are shared state, not personal state.

Task flows

F1 — PRIORITIZE THE MORNING TAKE (P1 DANA)

1. Open the workspace; the queue shows unworked threads by default.
2. Read new vs worked counts; toggle queue/all to see the whole corpus.
3. Scan content-type lanes — message, transcript, document — for where the volume sits.
4. Read facet counts as corpus-level signal before opening anything.
5. Apply scope chips to task the day; disposition markers record the team's decisions.

Exit: a work queue someone else can act on without asking what it means.

F2 — WORK A THREAD IN SCOPE (P2 MARISOL)

1. Filter to my-languages; the queue narrows to threads she is accountable for.
2. Open a thread. It renders bilingually — English leads, the source stays on screen.
3. Read the machine translation against the original, which is never off-screen.
4. Confirm, or correct against the source. The verdict badges the message.
5. Attach an analyst note where the judgment needs explaining.
6. Promote the thread when it is done. The stack ticks only for completed work.

Exit: a thread whose translation carries a named linguist's verdict downstream.

F3 — EXPLOIT THE SUBSTANCE (P3 KEN)

1. Search in content mode across originals *and* English translations.
2. A "matched in translation" badge distinguishes the two kinds of hit.
3. Select a hit; the viewer scrolls to the message and flashes it in context.
4. Request a summary for a long thread, on demand, collapsing it away after.
5. Open an attached document; OCR yields block-level text and an English gloss.
6. Tag and annotate; the tags re-enter triage as facets.

Exit: substance tied to its source, without a second tool.

F4 — SWEEP FOR SELECTORS AND ENTITIES (P4 PRIYA)

1. Switch to entity mode; restrict by type — person, geo, selector.
2. Apply a watchlist or geo-fence group as a one-click scope.
3. Narrow by date range.
4. Pivot from an entity chip to every other appearance.
5. Open the hit in its thread; provenance travels with anything exported.

Exit: a watchlist hit found this week rather than discovered in three.

Success criteria, and where each is verified

Every criterion below is replayed by a named specification. Task 5.2 records the result; `traceability_matrix.md` carries the same rows joined to the pain each one answers.

#	Persona	Success criterion	Verified by (spec file — test name)
S1	Dana	Content type, language scope and completion state are all readable from the queue	`queue.spec.js` — *content-type, language scope, and the stack ticks only when work is done*
S2	Dana	Progress reflects decisions taken, and the queue can be run clear	`queue.spec.js` — *queue progress ticks via strip decisions and the stack can run clear*
S3	Dana	Triage filters and group scopes return correctly scoped results	`search.spec.js` — *facet and group triage filters return scoped results*
S4	Dana	The find tools collapse away, leaving the queue as a work list	`queue-only.spec.js` — *find tools collapse to leave just the queue; stack controls stay*
S5	Marisol	A thread opens bilingually with the source always on screen	`bilingual.spec.js` — *threads open bilingual: English leads, source always on screen, no toggle*
S6	Marisol	Correction happens against the source, and review state is obvious	`bilingual.spec.js` — *correction happens against the source; review state is obvious*
S7	Marisol	RTL and mixed-script threads render without layout breakage	`viewer.spec.js` — *thread queue loads and Arabic message renders RTL*
S8	Marisol	A correction and an analyst note reach the output as written	`enrichment.spec.js` — *translation review and analyst note (smoke-6 review flow, clips removed)*
S9	Marisol	Enrichment is thread-level, not per-message hand-work	`enrichment.spec.js` — *auto-translation renders and thread-level entity extraction chips every message*
S10	Marisol	Reviewing a whole thread and promoting it ticks the stack	`ocr-threadgold.spec.js` — *thread gold: translate thread, review all, promote — the stack ticks*
S11	Ken	Search matches native script and English translation in one query	`viewer.spec.js` — *message search accepts native-script and cross-language queries*

#	Persona	Success criterion	Verified by (spec file — test name)
S12	Ken	Cross-language search holds across all four corpus languages	`search.spec.js` — *4-language search: native scripts, cross-language, stats*
S13	Ken	A hit lands on the evidence, in context	`viewer.spec.js` — *search hit lands on the evidence with a flash*
S14	Ken	Summaries are on demand and collapse away	`viewer.spec.js` — *summary widget renders on demand and collapses out of the way*
S15	Ken	Document text and an English gloss come out of an image, across pages	`multipage-doc.spec.js` — *multi-page Russian document: pages, Cyrillic blocks, and gloss*
S16	Ken	Extracted blocks name the engine that produced them	`ocr-threadgold.spec.js` — *OCR viewer: extracted blocks render with engine caption*
S17	Ken	Officer tags feed back into triage	`ocr-threadgold.spec.js` — *officer document annotation: tag + note, and the tag filters the queue*
S18	Priya	A group scope implies selector search, and the hit opens its thread	`search.spec.js` — *entity mode with a group implies selector search; hit opens the thread*
S19	Priya	A flagged-number sweep surfaces the contract	`multipage-doc.spec.js` — *the contract surfaces in a flagged-numbers watchlist sweep*
S20	All	One output: export is locked until a thread is promoted, then carries verdicts and notes	`enrichment.spec.js` — *the one output: export is locked until a thread is promoted, then carries verdicts and notes*

What changed during performance

Three findings altered this journey after the evaluated Phase 2 demonstration. Each is recorded as its own artifact with rationale and a restoration path:

- **The unit of gold is the thread, not a snippet** (`linguist_workflow_model.md`) — step 2 ends in a promoted thread rather than a tray of clips.
- **No verdict without the source** (`bilingual_display_model.md`) — step 2 reads bilingually by default; the source never leaves the screen.

- **A linguist’s bench, one output** (`one_output_model.md`) — the journey has a single exit, and the evidence-clip path was removed.

The third removes evaluated behavior. The React reference implementation in this repository is untouched and retains it, so strict parity is a single-commit revert.

Annotated Wireframes

UNCLASSIFIED — Annotated from the evaluated Phase 2 demo screens (see `../demo-state.png` for the captured reference state). Each annotation names the persona it serves (P1–P4, `personas.md`). The thread-gold additions from `linguist_workflow_model.md` are marked **[NEW — 31 Jul amendment]**.

Overall layout — three panels, independent collapse

```
┌──────────────────────────────────────────────────────────────────┐
│ ≡ IMAX · Triage Workspace   [UNCLASSIFIED — DEMONSTRATION DATA (FABRICATED)] │
← banner: yellow,
├─────────────┬───────────────────────────────┬──────────────────┤
center, always on
│ SEARCH &    │  CHAT LOG VIEWER              │  GOLD COPY       │
│ TRIAGE    ‹ │                               │              ›   │
│ (312px)     │  (flex)                       │  (320px)         │
│             │                               │                  │
│ collapse → 40px│  empty state until thread selected │  collapse → 40px rail │
│ rail, expand │                               │  rail keeps clip-count│
│ via nav-expand │                             │  badge (goldcopy- │
│             │                               │  expand)         │
└─────────────┴───────────────────────────────┴──────────────────┘
```

Collapse serves P2/P3 focus modes: a linguist reading RTL text collapses both side panels; an exploiter building product keeps Gold Copy pinned open.

Left panel — Search & Triage (P1 Dana, P2 Marisol, P4 Priya)

```
┌────────────────────────────────┐
│ SEARCH & TRIAGE    Clear ‹     │
│ My languages: [ar][fa][zh]     │ ← P2: sticky scope, "which of these are mine"
│                                │
│  ┌──────────────────────────┐  │
│  │ 🔍 query    [dir=auto]   │  │ ← RTL-safe input
│  └──────────────────────────┘  │
│ mode: (content|entity)         │ ← P3 content sweeps / P4 entity+selector hunts
│ types: msg○ trans○ doc○        │
│ facets: geo(12) person(9)      │ ← P1: corpus triage signal on every chip
│         passport(2) img(4)     │
│ groups: ▸Libya Coast           │ ← P4: geo-fence / watchlist = one-click scope
│         ▸Priority Handles      │
│ dates: [from]–[to]             │
├────────────────────────────────┤
│ QUEUE  (queue|all)  12/40      │ ← [NEW] N-of-M worked progress; stack-clear
│ ████████░░░░░░░░  worked       │   done-state when the stack empties
│ ✔ t-1001 Harbor Freight 📎     │ ← dispositions (✔ ⚑ ✖) = the stack's done marks;
│ ⚑ t-1002 Caravan Route         │   📎 = embedded doc cue
│   t-1003 Warehouse Manifest    │
└────────────────────────────────┘
```

Center panel — Chat Log Viewer (P2 Marisol, P3 Ken)

```
┌──────────────────────────────────────────────┐
│ Harbor Freight Coordination   GreenWire · 9  │
│ translated 6/9 · reviewed 4/9   [Translate   │ ← [NEW] gold-ready meter +
│                                  thread]     │   batch translate (context-
│                                              │   preserving unit of work)
│  ┌────────────────────────────────────────┐  │
│  │ saqr_92 · ar · 08:14              [RTL] │  │
│  │ وصلت الشحنة إلى الميناء صباح اليوم…      │  │ ← RTL/mixed-script correct
│  │ 🌐 The shipment arrived at the port…    │  │ ← translate-in-place
│  │ [✓ confirm] [✎ correct]  ✔ linguist-    │  │ ← P2: verdict travels with
│  │                           confirmed     │  │   the text into clips/gold
│  │ entities: [Abu Karim·person]            │  │ ← P4: chips, palette colors
│  │ 📝 note · 📎 bill_of_lading.png → OCR    │  │ ← P3: attachment → OCR dialog
│  │                             [clip ▸]    │  │ ← P3: snippet clip w/ prov.
│  └────────────────────────────────────────┘  │
│ [∑ Summarize]  scroll-to-hit + 1.8s flash    │
└──────────────────────────────────────────────┘
```

Right panel — Gold Copy (P3 Ken, P4 Priya; thread gold: P2 Marisol)

```
┌─────────────────────────────┐
│ GOLD COPY                 › │
│ ┌─────────────────────────┐ │
│ │ ▸ THREAD GOLD [NEW]     │ │ ← whole-thread unit: full transcript,
│ │  t-1001 9/9 tr · 9/9 rv│ │   original+translation, verdict badges,
│ │  [Promote to gold]      │ │   thread-level provenance header.
│ └─────────────────────────┘ │   Promotion ticks the stack (queue 13/40)
│ CLIPS (as evaluated)        │
│ ┌─────────────────────────┐ │
│ │ [ENTITY: person]        │ │ ← P3/P4 evidence tray, snippet-level
│ │ Abu Karim               │ │
│ │ thread t-1001 / msg m-3│ │ ← provenance line under every clip —
│ │ @harbor_ops / via mock │ │   traceability without bookkeeping
│ └─────────────────────────┘ │
│ [Export ▸]  (disabled when │ ← standardized product template render,
│             tray is empty) │   copy to clipboard
└─────────────────────────────┘
```

Annotation index (wireframe → requirement)

Annotation	Persona	Source
My-languages scope, dispositions, queue counts	P1, P2	Evaluated demo
Queue N-of-M progress + stack-clear	P1, P2	**[NEW]** `linguist_workflow_model.md`
Content/entity modes, facets, groups, dates	P3, P4	Evaluated demo
Translate-in-place + review verdicts	P2	Evaluated demo
Translate-thread + gold-ready meter	P2	**[NEW]** `linguist_workflow_model.md`
Snippet clips + provenance + export	P3, P4	Evaluated demo
Thread gold promotion + full-transcript export	P2	**[NEW]** `linguist_workflow_model.md`
RTL/mixed-script rendering	P2, P3	Evaluated demo
OCR dialog + officer annotation loop-back	P3	Evaluated demo

Pain-to-Design Traceability Matrix

UNCLASSIFIED — TDD subtask 2.4. Each persona pain point joined to the design response that answers it, the screen where that response lives, and the validation result that proves it works. Pains are drawn from two places in the research: **Frustrations today** in `personas.md` and **Pains removed** in `empathy_maps.md`. Validation is the executable specification that re-runs the claim, per Task 5.2.

This exists so “we addressed the research” is checkable rather than asserted. A row with no validation column is a design intention; a row with one is a delivered behaviour.

Reading it: *Screen* uses the three-panel vocabulary from `wireframes.md` — Left (Search & Triage), Center (Chat Log Viewer), Right (Gold Copy). *Evidence* names a screenshot in `project/evidence/screens/` showing the built state.

P1 — Dana, Collection Management Officer

Pain	Design response	Screen	Validation	Evidence
Cross-tool reconciliation of statistics against content	Counts and content on one screen: new/worked counts above the queue they describe	Left	`queue.spec.js` — *content-type, language scope, and the stack ticks only when work is done*	`day2-queue-only-view.png`
Opening threads just to find out whether they are worked	Disposition markers (reviewed / flagged / discarded) as shared team state, visible in the list	Left	`queue.spec.js` — *queue progress ticks via strip decisions and the stack can run clear*	`day2-review-states-strip.png`
No cheap way to tell an empty channel from a hot one	Facet counts as corpus-level triage signal, read before opening anything	Left	`search.spec.js` — *facet and group triage filters return scoped results*	`search-1-crosslang.png`
Volume without signal; tasking that is hard to defend	Content-type lanes (message / transcript / document) and queue/all toggle	Left	`queue-only.spec.js` — *find tools collapse to leave just the queue; stack controls stay*	`tour-2-working-stack.png`
Progress that counts activity rather than completion	The stack ticks only when a thread is actually finished, and can be run clear	Left	`queue.spec.js` — *queue progress ticks via strip decisions and the stack can run clear*	`tour-3-stack-clear.png`

P2 — Marisol, Language Officer

Pain	Design response	Screen	Validation	Evidence
Machine translation pasted onward loses its provenance	Verdict badges — linguist-confirmed / linguist-edited — that persist into the output	Center → Right	`enrichment.spec.js` — *translation review and analyst note (smoke-6 review flow, clips removed)*	`day1-bilingual-review.png`
Her correction does not travel with the text	One correction, made once, badges the message and lands in the export as written	Center → Right	`enrichment.spec.js` — *the one output: export is locked until a thread is promoted*	`day9-export-thread-gold.png`
Judging a translation without its source in view	Bilingual by default: English leads, the source line stays on screen, no toggle	Center	`bilingual.spec.js` — *threads open bilingual: English leads, source always on screen, no toggle*	`day1-bilingual-review.png`
Correcting against the translation rather than the original	Correction is written against the source; review state is visible while editing	Center	`bilingual.spec.js` — *correction happens against the source; review state is obvious*	`day2-review-states-strip.png`
RTL text breaks in generic viewers	RTL and mixed-script rendering in the thread and the queue	Center	`viewer.spec.js` — *thread queue loads and Arabic message renders RTL*	`day6-viewer-rtl-summary.png`
Re-translating already-worked messages	Translations render by default; worked state is visible before opening	Left, Center	`ocr-threadgold.spec.js` — *thread gold: translate thread, review all, promote — the stack ticks*	`day2-review-states-strip.png`

Pain	Design response	Screen	Validation	Evidence
Uncertainty about which threads are hers	My-languages scope chips as the only language scope	Left	`queue.spec.js` — *content-type, language scope, and the stack ticks only when work is done*	`day2-queue-only-view.png`

P3 — Ken, Subject-Matter Exploiter

Pain	Design response	Screen	Validation	Evidence
Search that stops at metadata	Content-mode search across originals and English translations in one query	Left	`viewer.spec.js` — *message search accepts native-script and cross-language queries*	`search-1-crosslang.png`
Cannot tell a source hit from a translation hit	”Matched in translation” badge distinguishes the two	Left	`search.spec.js` — *4-language search: native scripts, cross-language, stats*	`search-1-crosslang.png`
Losing thread context when quoting one message	Scroll-to-hit lands on the message and flashes it, in its thread	Center	`viewer.spec.js` — *search hit lands on the evidence with a flash*	`day6-hit-flash-landing.png`
Long threads that must be read end to end	Summary on demand, collapsing out of the way after	Center	`viewer.spec.js` — *summary widget renders on demand and collapses out of the way*	`day6-viewer-rtl-summary.png`
Attachments require a second tool	OCR viewer with block-level extraction and English gloss, multi-page	Center	`multipage-doc.spec.js` — *multi-page Russian document: pages, Cyrillic blocks, and gloss*	`day9-ocr-annotation.png`
Hand-built evidence trails	Officer tags and notes feed back into search and into the single output	Center → Right	`ocr-threadgold.spec.js` — *officer document annotation: tag + note, and the tag filters the queue*	`day9-thread-gold-promoted.png`
Any excerpt he cannot trace	Provenance line under everything that	Right	`enrichment.spec.js` — *the one output: export is*	`day9-export-thread-gold.png`

Pain	Design response	Screen	Validation	Evidence
	leaves the workspace		*locked until a thread is promoted*	

P4 — Priya, Targeting Officer

Pain	Design response	Screen	Validation	Evidence
Selector lists checked by hand against exports	Entity-mode search with type restriction (person / geo / selector)	Left	`search.spec.js` — *entity mode with a group implies selector search; hit opens the thread*	`search-2-watchlist.png`
A list checked weekly is checked too late	Watchlist and geo-fence groups as first-class one-click scopes	Left	`multipage-doc.spec.js` — *the contract surfaces in a flagged-numbers watchlist sweep*	`search-2-watchlist.png`
Geographic questions answered by keyword guesswork	Geo-fence group scope rather than free-text guessing	Left	`search.spec.js` — *facet and group triage filters return scoped results*	`search-2-watchlist.png`
Cannot pivot from a mention to every appearance	Entity chips with confidence, pivoting to all appearances; date-range narrowing	Left, Center	`search.spec.js` — *entity mode with a group implies selector search; hit opens the thread*	`tour-1-thread-open.png`
Exported entities lose their origin	Export carries the entity provenance line	Right	`enrichment.spec.js` — *the one output: export is locked until a thread is promoted*	`day9-export-thread-gold.png`

Coverage

Pain points traced (from `personas.md` **Frustrations today** and `empathy_maps.md` **Pains removed**)	24
Traced to a design response and a screen	24 (100%)
Carrying an executable validation result	24 (100%)
Carrying a built-state screenshot	24 (100%)
Distinct specifications cited	8

Every pain traces to something that runs. That is the claim this document exists to make checkable — and the reason each row names a specification rather than a paragraph.

The eight specifications cited here are not the whole suite: `tests/ng/` holds sixteen files. This document traces the persona pains identified at design time, so it cites the specifications that validate the design responses to them. The other eight cover work that arrived later during performance — the twelve internal-review defects, the officer's document-translation verdicts, and the virtualized message stream TDD 3.3 commits to — and are accounted for in D3, not here. A row is added to this matrix only when a persona pain is what motivated the change.

Two rows are approximate rather than exact matches, and are marked here rather than left for a reader to notice: *"Volume without signal; tasking that is hard to defend"* is evidenced by the queue-only view rather than by a test of the lanes themselves, and *"Re-translating already-worked messages"* is evidenced by the promote-the-thread flow rather than by a test that re-opening a worked thread skips translation. Both behaviours exist; neither has a specification aimed squarely at it.

Caveats, stated rather than smoothed

- **Personas are composites, not interviews.** Direct end-user access was not assumed and was not granted; personas derive from the Phase 2 solution demonstration and its evaluated workflows. Everything they assert is an inference from that material, and the assumptions carrying the most weight are logged in `assumption_log.md`.
- **Validation is against fabricated demonstration data.** The specifications prove the behaviour, not that the behaviour is right for real traffic at real volume.

- **Three rows describe behaviour that changed during performance.** The bilingual default, thread-level gold, and the single output path each replaced an evaluated behaviour; the rationale and restoration path are in the three workflow-model artifacts.
- **Not validated with the Sponsor.** Touchpoint #1 had not occurred at time of delivery. These are findings offered for validation, not findings the Government has accepted.

Assumption Log

UNCLASSIFIED — TDD subtask 2.4. What we assumed, why we had to, what it would cost if the assumption is wrong, and how to find out.

Direct end-user access was not assumed in the TDD and was not granted during performance. Personas are therefore composites inferred from the Phase 2 solution demonstration and its evaluated workflows. That is a defensible method on a fourteen-day prototype, but it means the design rests on stated assumptions rather than observed users — and an assumption that is written down can be checked, while one that is merely held cannot.

Status is as at delivery. **Open** means nobody has confirmed it.

Design assumptions

#	Assumption	Why we made it	If wrong	How to close it	Status
A1	The linguist's unit of work is the whole thread, not a snippet	Persona work with Marisol's flow: a correction made on one message is meaningless without the surrounding exchange	The thread-level gold flow is the wrong shape; snippet clipping returns	Touchpoint #1 — show a linguist the promoted-thread flow and watch where they hesitate	**Open** — offered, not validated
A2	A native speaker will not render judgment without the source in view	A translation reviewed against itself is unreviewable	The bilingual default wastes vertical space for no gain	Touchpoint #1 — ask whether they ever want translation-only	**Open** — offered, not validated
A3	This application is a linguist's bench, and other roles consume its output downstream	Four personas fit in one screen only by making one of them the operator	Ken and Priya need first-class production paths, not read paths	Touchpoint #1 — ask who is expected to sit in front of it daily	**Open** — logged deviation #2
A4	Disposition state is shared across the team, not personal	Dana's tasking is defensible only if the markers mean the same thing to everyone	Needs per-user state and a merge story	Ask whether two officers may disagree about a thread's disposition	**Open**
A5	Four personas bracket the real user population	The evaluated workflows describe these four roles; more would not have fitted the sprint	A significant workflow is unrepresented	Touchpoint #1 — ask which role is missing	**Open**
A6	Machine translation is close enough to be worth	The whole confirm/correct flow presumes it	The verdict badges describe an interaction nobody performs	First contact with the enclave gateway on	**Open** — gateway unproven

#	Assumption	Why we made it	If wrong	How to close it	Status
	correcting rather than replacing			real language pairs	

Environment assumptions

#	Assumption	Why we made it	If wrong	How to close it	Status
A7	The authenticating front is the only path to the pod	Identity headers are trusted absolutely; nothing else makes that safe	Any caller can assert any identity, including group membership	Confirmed by River Hawk, 2 Aug	**Closed**
A8	The front forwards identity as an STS token, not flat headers	Stated by the Sponsor, 3 Aug: `Authorization` carries the identity token	Every request is refused behind the real front	Implemented as `AUTH_MODE=bearer-jwt`; unverified against a real token	**Partly closed** — mechanism confirmed, claim names not
A9	The STS token's claims use the common OIDC spellings (`sub`, `name`, `groups`, `org`)	No claim inventory was provided	Identity resolves empty and every request 401s	`GET /api/whoami` reports the claim names a real token carries; the fix is configuration	**Open**
A10	Server certificates arrive via AWS Secrets Manager	Sponsor said certs are provided up front with no clear direction on how; the environment is AWS C2S	The initContainer path is unused; the file-based overlay is used instead	Both paths are built and proven; ask which	**Open** — both supported
A11	Fixtures are an acceptable fallback when the model gateway is unreachable	Confirmed by the Sponsor: test data is fine	A demonstration silently shows canned output	Confirmed 3 Aug; pods now log which mode they are in	**Closed**

#	Assumption	Why we made it	If wrong	How to close it	Status
A12	The cluster runs images we built, or rebuilds them from our source	Both are supported; which one is a platform policy question	A ferried image set is rejected, or a rebuild stalls on a missing mirror package	Ask whether the pipeline builds from source against an approved base	**Open**

Process assumptions

#	Assumption	Why we made it	If wrong	How to close it	Status
A13	Angular v21 LTS, pinned	Stated in writing at kickoff, unanswered	Rework against a different major	Confirmed by the Sponsor, 3 Aug: Angular 21	**Closed**
A14	Deliver-deployable rather than deploy-for-us	Kickoff assumption stated in writing	The delivery posture is wrong	Sponsor asked for a high-side working demo on their system, 3 Aug	**Closed** — deploy into their sandbox
A15	Sandbox access would arrive during performance	Required for the deployment claim	Deployment is validated on our own cluster instead — which is what happened	Access in progress; onboarding session pending	**Open** — the reason DELIVERY.md does not claim Platforma
A16	Wireframes could be frozen at Touchpoint 2 (about Day 7)	TDD schedule	Design churn into the build window	Touchpoints did not occur on the TDD schedule; wireframes were frozen against the evaluated demonstration instead	**Closed by substitution**

What this log implies

Six of the sixteen assumptions are closed. Of the ten still open, **five are design assumptions awaiting Touchpoint #1** (A1–A5) — the prototype embodies them, and the whole purpose of the touchpoint is to find out whether they hold. The rest are environment questions that resolve on first contact with Platforma.

None of them is a defect. An assumption is only dangerous when it is invisible, and the reason this document exists is that the TDD promised it and a design built on inference rather than interviews needs somewhere to say so plainly.

Workflow Model: The Linguist's Finite Stack

UNCLASSIFIED — HCD finding and design response, recorded 31 July 2026. Status: **adopted for the Phase 3 build (additive)**; to be validated with the Sponsor at touchpoint #1 (Mon 3 Aug).

The finding

A Language Officer's work is a **finite stack**: a queue of threads and documents in their languages, worked one at a time, each one *done when it's done* — fully translated and reviewed — until the stack is clear. This is the working model of every mature translation workflow (CAT/TMS tools are built on it), and it is how linguists describe their own day.

The Phase 2 demo already embodies half of this model: the triage queue with My-languages scope, new/worked counts, and dispositions (reviewed / flagged / discarded) *is* the stack. What it gets wrong is the **granularity of the output**. The Gold Copy pane collects snippets — one message, one entity, one OCR gloss at a time. That is an *analyst's* evidence tray (personas P3 Ken and P4 Priya, see `personas.md`), and it is the right tool for building an intelligence product from worked material. But it is not what a linguist produces. A linguist's gold copy is the **entire translated thread or document** — because translation quality lives in context. A sentence translated in isolation loses referents, register, and continuity; the same reason machine translation of a whole document beats sentence-by-sentence translation. Clipping snippet-by-snippet strips exactly the context that makes the linguist's judgment trustworthy.

Principle: the unit of gold must match the unit of context. For the linguist, that unit is the thread.

Persona mapping

Persona	Unit of work	Unit of output	Served today by
P2 Marisol (Language Officer)	Thread from her stack	**Whole-thread gold copy** (full transcript, translations, her verdicts)	*Gap — closed by this amendment*
P3 Ken (SME exploiter)	Evidence trail across threads	Snippet clips with provenance	Gold Copy clip tray (as evaluated)
P4 Priya (Targeting Officer)	Selector/entity hits	Entity clips with provenance	Gold Copy clip tray (as evaluated)
P1 Dana (CMO)	The stack itself	Queue state (worked/unworked counts)	Triage queue + dispositions

Design response (additive — the evaluated clip flow is unchanged)

1. **Translate thread** — batch translate-in-place across every message in the thread (prototype iterates the per-message mock service; the production seam would pass full-thread context to the translation service, per the principle above).
2. **Thread review completion** — the thread header shows `translated n/m · reviewed n/m`; a thread is *gold-ready* when every message carries a linguist verdict (linguist-confirmed or linguist-edited).
3. **Promote thread to Gold** — one action producing a thread-level gold copy: full transcript (original + translation side by side), verdict badges, and a thread-level provenance header (thread, network, participants, date range, services, timestamps). Rendered in the Gold Copy pane above the clip tray; exportable through the existing export dialog as a full-translation product.
4. **The stack ticks down** — promoting a thread to gold marks it worked; the queue shows `N of M worked` progress and an explicit stack-clear done-state. “When it’s done, it’s done” becomes visible.

New testids (additions only, no renames): `thread-translate-all`, `thread-gold-ready`, `promote-thread-gold`, `thread-gold-*`, `queue-progress`, `stack-clear`.

Update (1 Aug) — the stack only marks work done

Design review sharpened the disposition semantics: **a linguist does not discard work from the queue.** The stack is assigned work; the only queue-side action is *reviewed* (worked). Flag-to-targeter and discard are thread-level judgments that require having read the thread, so they live in the viewer's workflow strip — the same single place as promote-to-gold. All thread decisions in one strip; the queue stays a stack. The gold pane also slimmed (280px) — it is an output surface, not a working surface — and collapses to a rail. Queue-side flag/discard testids moved to the strip unchanged; the ported acceptance suite only ever used the queue's reviewed action, so smoke-5 is untouched.

Contract posture

The evaluated Phase 2 baseline (three panels, snippet clips, export) ships with visual parity. This model is an HCD-driven **addition** (~1 person-day), presented to the Sponsor at touchpoint #1 as a finding from persona work, and logged in the acceptance package. If the Sponsor prefers strict parity, the addition is cleanly removable — it touches no evaluated behavior.

Workflow Model: No Verdict Without the Source

UNCLASSIFIED — HCD finding and design response, recorded 31 July 2026. Status: **adopted for the Phase 3 build (additive)**; to be validated with the Sponsor at touchpoint #1 (Mon 3 Aug) alongside `linguist_workflow_model.md`.

The finding

The evaluated demo's translate-in-place is a **swap**: the English replaces the original, and an undo button brings the original back. That is the right ergonomics for a *reader* — an exploiter who wants the thread in English. It is the wrong surface for a *judge*. The linguist's core act in this tool is a review verdict — confirm or correct the machine translation — and in the swap model that verdict is rendered **while the source text is hidden**. A native speaker never signs off on a translation they cannot see the original of; every professional translation environment (CAT tools, review benches) shows **bitext** — source and target together — for exactly this reason.

This is the same principle family as `linguist_workflow_model.md`: translation judgment lives in context. That artifact fixed the granularity of the *output* (the unit of gold is the thread); this one fixes the granularity of the *view* (the unit of judgment is the source-target pair).

Principle: no verdict without the source. Whenever a translation is displayed — reading, confirming, or correcting — the original stays on screen with it.

Persona mapping

Persona	Swap view (as evaluated)	Bilingual view (this finding)
P2 Marisol (Language Officer)	Verdicts rendered blind to the source	Confirms/corrects against the source line every time
P3 Ken (SME exploiter)	Reads in English — fine	Unchanged: English leads, source is a quiet secondary line
P4 Priya (Targeting Officer)	Selector strings can differ between source and MT	Sees the original selector string alongside the English

Design response (additive — the evaluated swap semantics are unchanged)

1. **Translated view becomes bilingual.** When a message shows its English (machine or linguist version), the **original stays visible beneath it** as a quoted secondary line in its own script and direction. The English leads (reader ergonomics preserved); the source never leaves the screen (judge ergonomics gained).
2. **Correction happens against the source.** The correction editor renders with the original directly above the draft — the linguist edits while reading the source, not from memory.
3. Thread-gold transcripts already comply (`ORIG:` / `EN [verdict]:` per message) — this finding brings the live reading surface in line with the gold output.

New testid (addition only, no renames): `original-<messageId>` on the persistent source line. *(As first recorded on 31 Jul this response kept the evaluated show/hide toggle; the 1 Aug update below removed it — bilingual is the only translated view.)*

Update (1 Aug) — translations render by default, and review state is visible

Three follow-on decisions from design review:

1. **Auto-render the baseline translation — and drop the toggle.** Machine translation is cheap; making the linguist click per message (or per thread) to see it was friction without judgment value. Threads now open in the bilingual view with the baseline MT already rendered, and the per-message show/hide toggle was removed outright: the source is always on screen, so “show original” had nothing left to show. **Tiered translation is the production seam**: the auto-rendered baseline is the fast tier; a premium frontier-LLM retranslation on demand (per message or per thread) is the documented cutover behavior — not mocked in the prototype, named in the developer guide’s production-cutover note.
2. **Review state at a glance.** Each message carries a colored edge: green (linguist-confirmed), amber (linguist-edited), dashed grey (translated, awaiting review). A thread’s progress toward gold-ready is scannable without reading badges.
3. **”Gold” is reserved for promotion.** There is exactly one gold action — promote the thread from the viewer strip. Snippet clipping remains (exploiter/targeting workflow) but is labeled **evidence**, with its own tray section, so the two outputs can’t be confused.

(Decision 3’s “evidence” relabeling was superseded the same day by `one_output_model.md`, which removed snippet clipping from the bench entirely.)

Contract posture

The 31 Jul source line was additive; the 1 Aug default-on + toggle-removal steps change evaluated behavior (`translate-*` per-message controls no longer exist in the Angular build). Logged with `one_output_model.md` as the linguist-bench deviation set: ported specs 2/6/7 amended, 1/3/4/5 verbatim, React reference app untouched, single-commit restoration path if the Sponsor requires strict parity.

Workflow Model: A Linguist's Bench, One Output

UNCLASSIFIED — HCD finding and design response, recorded 1 August 2026. Status: **adopted for the Phase 3 build**; on the touchpoint-#1 agenda (Mon 3 Aug) with the two 31 July findings. This artifact supersedes the "evidence clips" terminology in the 1 Aug updates of `linguist_workflow_model.md`.

The finding

The evaluated demo was a multi-role console: linguist review, analyst evidence-clipping, and targeting sweeps sharing one screen, with clip scissors on every message and two competing output paths (snippet tray vs. promoted thread). Design review asked a simple question — *"what does clip-to-evidence do?"* — and the honest answer exposed the problem: the workspace's primary user could not tell, because it wasn't for them.

This application is a linguist's bench. The linguist works a finite stack, judges translations against the source, and produces exactly one thing: the **gold copy of the whole thread** — full transcript, verdicts, analyst notes. Everything else that other roles need is a *downstream consumption* of that output or of the linguist's hand-offs:

Downstream need	Served by
Targeting follow-up	**Flag for targeter** (thread strip) + entity chips as spotting aids
Product building (exploiter)	**Gold copy export** — full verdicted transcripts with NOTE lines
Supervision (CMO)	Queue counts, N-of-M progress, review edges

Design response

1. **Evidence clips removed.** No scissors anywhere; the gold pane holds promoted thread gold only; export renders full translations. Analyst notes survive — they annotate the translation and travel into the gold transcript as `NOTE [analyst]:` lines.
2. **Enrichment is thread-level, once.** Entity extraction joins translate-thread in the workflow strip: one action for the whole thread (the same context principle — the unit of enrichment is the unit of work). Chips render on messages as display-only spotting aids.

3. **The stack has no buttons.** Queue rows carry no actions at all. A queue-side “mark reviewed” let a linguist declare work done *without doing it* — the thread wasn’t translated, judged, or promoted. **Done means promoted to gold**, or a deliberate flag/discard taken in the viewer strip after reading. Rows still show state (reviewed / flagged badges, strikethrough on discarded); they just don’t accept decisions.
4. **Sort options match a work queue.** *Untranslated first* died with auto-translation (everything arrives translated). The queue now defaults to **Oldest first** — a stack is worked in arrival order — with *Most recent* and *Message count* as alternates.
5. **Find tools collapse; the stack stays.** The panel separates *stack controls* (My languages, sort — properties of the queue) from *find tools* (search, content/ entity modes, content-type lanes, date range, facets, geo-fence/watchlist groups, officer tags — discovery). One toggle in the panel header hides the find tools, leaving a queue-only bench: the stack goes from five visible rows to fourteen. Hiding also clears any active query, facet, group, or tag filter — “hide search” always returns the linguist to their own stack, never to a filtered subset. Default is expanded (search remains a first-class capability, and the ported search acceptance flows run untouched); the collapsed state is one click away. **This is also the preview surface** if Sponsor feedback directs removing corpus search from the linguist build: the queue-only view *is* that product, so the trim can be evaluated before it is committed to.
6. Combined with the prior findings, the bench now reads in one sentence: *open a thread from your stack; it arrives translated and bilingual; judge every message against the source; promote to gold; the stack ticks down.*

Contract posture (logged deviation #2)

This removes evaluated behavior (snippet clip → tray → product export), a larger step than the additive findings. Logged accordingly:

- The React reference app is untouched and retains the full evaluated clip flow.
- Ported acceptance specs 2, 5, 6, and 7 are amended: clip/export beats removed and smoke-5’s queue-side disposition beat replaced by the real bench path (review the thread → promote → the stack ticks); review / entity / annotation / tag-facet / scope / content-type beats retained. Specs 1, 3, 4 remain verbatim.
- Restoration path if the Sponsor requires strict parity: the clip layer was removed in a single commit and reverts cleanly.
- Sponsor validation: touchpoint #1 presents the bench model with the parity fallback explicit.